# Myntra — Cloud Scaling for Flash Sales and Personalization

Week 2 Comprehensive Deliverable

Case Study Analysis, Architecture, Exercise Plan and Team Roles

**Course: Cloud Computing | Team Number: 33 | Date: November 2026**


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# 1. Introduction

## 1.1 Company Background

Myntra is India's premier fashion, beauty, and lifestyle e-commerce platform, founded in 2007 and acquired by Flipkart in 2014 (part of the Walmart group). The platform manages a catalog of more than 700,000 styles from over 3,000 international and domestic brands, serving tens of millions of active shoppers across India [V1].

Historically, Myntra operated on an on-premise monolithic architecture built upon PHP and MySQL. While adequate for early steady-state traffic, this monolithic stack suffered catastrophic bottlenecks during promotional spikes, characterized by database connection exhaustion, CPU saturation, long provisioning lead times, and single-point-of-failure vulnerabilities. To support exponential commercial growth, Myntra engineered a multi-year transition toward a decoupled, event-driven, microservice-oriented cloud architecture hosted on Microsoft Azure [V1].

## 1.2 Project Motivation & Case-Study Objectives

This case study investigates how hyperscale e-commerce platforms tackle two intersecting cloud computing imperatives:

1. Elastic Scalability under Flash-Sale Surges: Absorbing abrupt 15–20× traffic surges during marquee events such as the End of Reason Sale (EORS) and Big Fashion Festival (BFF) without over-provisioning idle, costly capacity during baseline windows.

2. Real-Time Personalization at Petabyte Scale: Concurrently capturing tens of billions of clickstream interactions, processing near-real-time behavioral features with second-level freshness, and executing ML inference within strict sub-millisecond budgets.

The overarching objectives of this project are: (a) conduct a rigorous, evidence-based architectural case study of Myntra; (b) analyze the failure modes of fixed infrastructure; (c) formulate a scaled-down practical simulation ("MyntraFlash") on Google Cloud Platform (GCP); and (d) establish a reproducible empirical methodology to quantify latency, horizontal elasticity, fault recovery, and cost efficiency in Week 3.

## 1.3 Scope and Research Boundaries

This Week 2 deliverable encompasses comprehensive case analysis, architectural modeling, experimental design, and governance. Per academic integrity principles, all production metrics cited herein are strictly attributed to verified primary sources (Myntra Engineering blogs, Microsoft Azure case studies, Aerospike technical reports, and Databricks publications). All metrics for the proposed university implementation represent planned hypotheses, target SLA thresholds, and verified GCP billing calculator estimates, clearly demarcated from measured experimental results.

## 2. Problem Statement

### 2.1 The Flash-Sale Concurrency Challenge

Flash sales such as Myntra's End of Reason Sale (EORS) represent an extreme class of cloud workloads. Unlike organic daily traffic that follows predictable diurnal patterns, flash-sale demand exhibits near-instantaneous step-function spikes:

| | | | |
|---|---|---|---|
| **EORS Traffic Spike Multiplier** | 15× to 20× normal daily load | Myntra Engineering (BCP/DR) [P3] | 2020–2021 |
| **EORS Order Velocity Target** | 12,000 orders/minute (10th/11th EORS) | Myntra Engineering / Retail Reports [P3] | 2019 |
| **EORS Order Velocity Capacity** | 20,000 orders/minute peak concurrency | Myntra Corporate Press Release [P3] | 2020 |
| **BFF Customer Visits** | 460 Million visits (2023) → 627 Million (2024) | Myntra Post-Sale Official Reports | 2023–2024 |
| **Marquee Day-1 Telemetry** | 18.8 Billion app/web events on Day 1 | Myntra Engineering (MDP) [P1] | 2022 |
| **Data Serving Volume** | 2.5 Petabytes (PB) / day served during BFF | Myntra Engineering (Bifrost) [P1] | 2022 |
| **Analytical Query Load** | 65,000+ queries/day across internal BI/services | Myntra Engineering (Data Platform) [P1] | 2022 |

### 2.2 Deep Subsystem Bottlenecks

Under flash-sale peak load, bottlenecks emerge across multiple architectural layers:

• Pricing & Discount Latency: The Product Listing Page (PLP) displays up to 50 items simultaneously. If discount evaluation takes 150 ms per item or makes sequential calls, total page render latency exceeds 400 ms, triggering customer abandonment [P2, P2b].

• Cart Service Tombstone Accumulation: During intense checkout windows, rapid additions, deletions, and updates (35% of cart ops) on Apache Cassandra generated severe tombstone accumulation, driving read latencies >250 ms with 0.05% request drops [P8].

• Online Feature Retrieval Ceilings: Real-time ML models require hundreds of user and item features per pageview. Legacy Redis clusters suffered lookup latencies of 8.5 ms, creating an unacceptable latency tax when servicing 500,000 ops/second [V2].

• Binlog Contention during Ingestion: Traditional batch ETL polling production MySQL masters degraded transaction throughput. Ingestion frameworks had to scale from a baseline of ~4 million events/min to peak spikes of up to 42 million events/min [P4, P7].

### 2.3 Infrastructure Dilemma: Peak Over-Provisioning vs. Elastic Scaling

Provisioning physical or fixed cloud infrastructure for 20× peak capacity results in >85% idle resource waste across 350 off-peak days, imposing an unsustainable capital and operational expenditure. Conversely, under-provisioning guarantees service crashes, lost revenue, and brand impairment during marquee sales. The engineering imperative is dynamic horizontal elasticity backed by rapid self-healing.

## 3. Research Findings & Empirical Benchmarks

### 3.1 Verified Primary-Source Metrics

The following table consolidates verified outcomes documented by Myntra Engineering and its cloud/database technology partners:

| | | | | |
|---|---|---|---|---|
| **Pricing & Discount Service** | ~150 ms latency | ≤ 30 ms (5× speedup) | Redis read balancing + In-Memory Rule Executor | [P2] |
| **Cart Persistence Service** | > 250 ms read latency | < 25 ms (10× speedup) | Cassandra to Sharded MongoDB migration | [P8] |
| **Personalization Feature Store** | 8.5 ms lookup latency | 0.8 ms (10× speedup) | Aerospike Hybrid Memory Architecture (NVMe+RAM) | [V2] |
| **Feature Store Concurrency** | Standard throughput | 500,000 ops/second | Aerospike Enterprise clustering on flash storage | [V2] |
| **Clickstream Ingestion Pipeline** | ~4M events/min (base) | 42M events/min (peak) | Meterial pipeline: Go collectors + Spark + Kafka | [P7] |
| **Enterprise Data Platform** | Legacy Hive locking | 35% Cost Reduction | Databricks Delta Lake unified lakehouse | [V3] |
| **Streaming Pipeline Processing** | Legacy ETL lag | 25% Faster Pipelines | Kappa architecture with Delta Lake & Spark | [V3] |
| **Marquee EORS Platform Scale** | 12K orders/min (2019) | 20K orders/min (2020) | Azure Kubernetes Service (AKS) + Azure Cloud | [V1, P3] |
| **Supply Chain BCP/DR** | Cold standby idle VMs | 200+ microservices live | Active utilization of DR Region B for SCM workloads | [P3] |

### 3.2 Critical Source Evaluation & Methodological Caveats

1. Temporal Disjunction: The cited metrics span 2019 to 2024. They reflect continuous iterative modernization rather than a single monolithic cutover. For example, the Azure cloud migration occurred in early 2020 [V1], the BCP/DR SCM migration was documented in late 2021 [P3], the Quicksilver real-time platform in 2022 [P5], and the Bifrost II Delta Lake lakehouse in 2024 [P1b].

2. Workload-Specific Improvements: Databricks' reported 35% cost savings pertains strictly to data engineering and analytics pipelines, not Myntra's holistic Azure infrastructure bill [V3]. Similarly, Aerospike's 0.8 ms latency describes online ML feature lookups, distinct from end-to-end HTTP round-trip times [V2].

3. Attribution Discipline: While Microsoft reported a 50% increase in EORS orders following the 2020 Azure migration [V1], this coincided with broader macroeconomic e-commerce adoption during COVID-19 and platform catalog expansion; it cannot be attributed solely to cloud infrastructure.

## 4. Myntra's Researched Cloud Architecture

### 4.1 Microsoft Azure Migration & Microservice Topology

In February 2020, Myntra announced a strategic partnership with Microsoft to migrate its core platforms to Microsoft Azure [V1]. The operational topology relies on Azure Kubernetes Service (AKS) for containerized application microservices, combining Horizontal Pod Autoscalers (HPA) for application tiers and underlying node pool expansion. To circumvent DNS resolution bottlenecks during sudden pod scale-outs, Myntra deployed NodeLocal DNSCache across AKS worker nodes.

### 4.2 High Availability, BCP and Disaster Recovery Topology

Myntra's Business Continuity Plan (BCP) and Disaster Recovery (DR) architecture operates across two Azure geographical regions [P3]:

• Active Dual-Region Resource Utilization: Rather than maintaining Region B as an expensive, passive cold standby, Myntra migrated over 200 microservices comprising its Supply Chain Management (SCM) ecosystem to run live production traffic out of Region B [P3]. Storefront browsing and order-capture paths reside in Region A, with seamless failover capabilities.

• Active-Active Messaging via Airbus: Myntra's internal messaging platform ("Airbus"), engineered on Apache Kafka, maintains bidirectional cross-region topic replication, ensuring at-least-once inter-service event synchronization [P3].

• Isolated Regional Observability: Logging and APM telemetry stacks are deployed as completely disjoint, independent clusters in each region, preventing cross-region egress overhead and eliminating shared failure dependencies [P3].

### 4.3 Production Architecture Conceptual Diagram

*The architecture diagram below synthesizes the verified relationships across Myntra's customer-facing tiers, low-latency pricing caches, Sourcerer ingestion, Kafka event bus, Quicksilver stream processing, Bifrost lakehouse, and Aerospike feature store.*

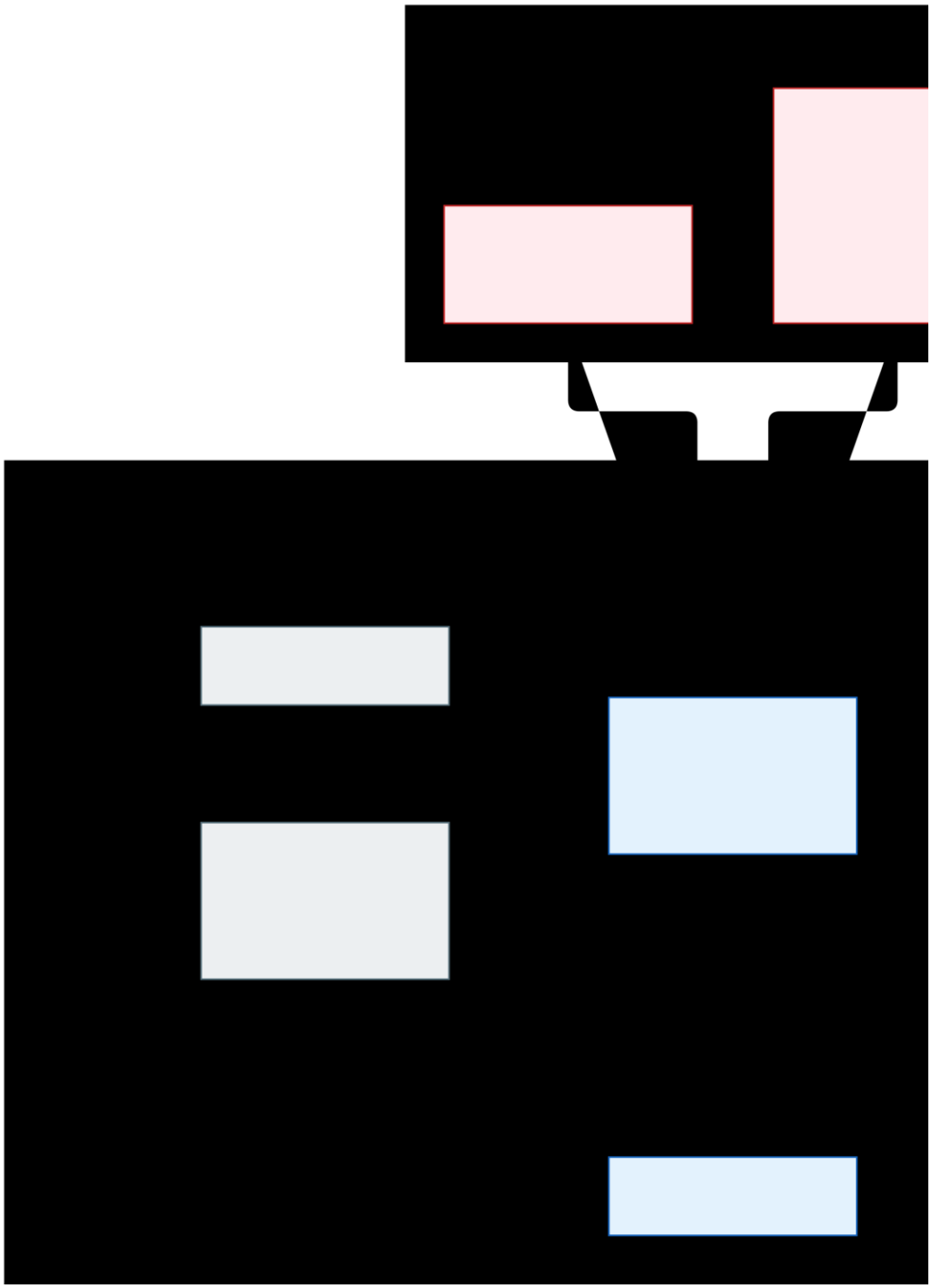


*Figure 1: Conceptual reconstruction of publicly documented Myntra architecture.*

## 5. Data Architecture & Real-Time Personalization

### 5.1 Sourcerer: CDC Event Ingestion Framework

Documented by Abhinav Dangi [P4], Sourcerer replaced brittle batch ETL with Change Data Capture (CDC) utilizing Debezium on Kafka Connect. To protect operational transactional databases (MySQL), dedicated read-replicas ("analytical read-slaves") are provisioned exclusively for binlog extraction. Sourcerer serializes database row mutations into Apache Avro, registers schemas with a centralized Schema Registry, and streams events into domain-partitioned Kafka topics with zero production contention.

### 5.2 Quicksilver: Near-Real-Time Stream Processing

Authored by Aditya S. [P5], Quicksilver handles in-session customer hyper-personalization. Built on Apache Spark Streaming, Quicksilver processes tens of billions of events daily with end-to-end freshness latencies of a few seconds. It maintains stateful stream windows spanning days to months, computing dynamic user engagement scores (e.g., brand affinity, recent price sensitivity) to update homepage feeds and product rankings in real time.

### 5.3 Janus: Multi-Engine Pipeline Orchestration

Engineered by Kedar Gupte [P6], Janus functions as Myntra's unified data orchestration framework. It provides low-code pipeline configuration, automated CI/CD deployments across multi-engine runtimes (Spark batch and streaming), and rigorous Directed Acyclic Graph (DAG) dependency management. Janus prevents data corruption and eliminates expensive downstream backfills by enforcing strict upstream dataset validation before triggering dependent processing stages.

### 5.4 Bifrost & Bifrost II: Unified Lakehouse Serving Layer

As detailed by Abhinav Dangi [P1b] and Sri Nandhini [P1], Bifrost serves analytical and reporting queries across 2.5 PB/day of data. Phase 3 decoupled compute from storage using Trino (formerly PrestoSQL) querying Hive ACID tables. Phase 4 ("Bifrost II") migrated storage to Databricks Delta Lake, establishing a Medallion architecture (Bronze raw, Silver cleaned/joined, Gold aggregated) that eliminated file-locking deadlocks, solved small-file metadata bloat, and unified batch and streaming tables via MVCC.

### 5.5 Aerospike Feature Store & ML Personalization Pipeline

For online ML inference during page loads, Myntra deploys Aerospike Enterprise Edition as its low-latency online feature store [V2]. Aerospike's Hybrid Memory Architecture (HMA) maintains primary indices in RAM while writing feature values directly to high-performance NVMe flash storage. This architecture slashed p99 feature retrieval latencies from 8.5 ms to 0.8 ms while scaling to 500,000 operations/sec, powering dynamic widget ranking and conversational memory for GenAI assistants (MyFashionGPT, Maya) [V1, V2].

## 6. Technical Case Analysis & Engineering Trade-Offs

### 6.1 Pricing System Caching & Read Balancing

Myntra's pricing system optimization [P2] provides a textbook case in low-latency distributed caching:

• Segregation of Browsing vs. Checkout Paths: Browsing represents >99% of pricing queries. Promotional rules, category coupons, and bank offers are pre-computed and pre-bundled into in-memory price structures, decoupling browsing reads from transactional DBs.

• Scalable Rule Executor Service (RES): Evaluates complex discount combinations entirely in memory. Rules are partitioned across memory shards keyed by SellerId, BrandId, or ProductId, enforcing strict shard capacity limits to eliminate CPU hotspots [P2].

• Redis Read Balancing & Cache Stampede Defense: Read queries are load-balanced across Redis read-replicas. Cache stampedes (where simultaneous cache misses overwhelm the database) are prevented through probabilistic early expiration and background mutex recomputation locks.

• Resilient Degradation: If Redis clusters experience transient latency spikes, the system serves cached pricing via Stale-While-Revalidate (SWR) semantics, strictly enforcing an upper SLA ceiling of 30 ms.

### 6.2 Storage Architecture Evolution: Cassandra vs. MongoDB

As documented in [P8], Myntra's shopping cart migration from Cassandra to MongoDB illustrates how workload access patterns dictate database selection. Cassandra's LSM-tree architecture excels at immutable write-heavy streams but degrades under frequent row deletions and updates due to tombstone accumulation during compactions. In contrast, MongoDB's B-tree document model handles in-place updates and atomic array modifications efficiently, reducing cart read latency from >250 ms to <25 ms with 40% less hardware.

### 6.3 Resilience Patterns: Circuit Breakers, Bulkheads & Rate Limiting

During flash sales, service failures cascade rapidly if unconstrained. Myntra isolates services via: (1) Bulkheading: Thread pools and socket connections for downstream services (e.g., recommendation engine) are strictly isolated so that downstream stalls never exhaust front-end web worker threads; (2) Circuit Breakers: Repeated timeouts trip circuit breakers, instantly reverting to cached static catalog fallbacks; and (3) Token-Bucket Rate Limiting: Safeguards checkout checkout APIs against bot traffic.

# 7. Proposed Practical Solution: MyntraFlash (Google Cloud Platform)

## 7.1 Simulation Concept & Cloud Provider Selection

To demonstrate the core principles of horizontal elasticity, load balancing, health monitoring, and cost efficiency in a controlled university laboratory environment, we have designed **MyntraFlash** — a scaled-down, cloud-native e-commerce simulation deployed on **Google Cloud Platform (GCP)**.

Following a rigorous review of student cloud grant models, Google Cloud Platform was selected for the practical exercise. GCP provides **$300 USD in free trial credits over 90 days** accessible to any standard developer account without requiring institutional college email whitelisting. Furthermore, GCP enforces a strict **zero auto-charge guarantee** (services pause when trial credits expire unless explicitly upgraded), eliminating any risk of out-of-pocket charges while delivering 100% architectural equivalence to enterprise cloud scaling systems.

## 7.2 Architectural Components & GCP Technology Stack

| | | |
|---|---|---|
| **Cloud Provider** | Google Cloud Platform (GCP) | asia-south1 (Mumbai) region; $300 free trial credits (90 days); zero auto-charge guarantee |
| **Compute Infrastructure** | GCE Managed Instance Groups (MIG) | Zonal/Regional managed pool of identical compute instances; direct equivalent to Azure VMSS |
| **Compute Machine Sizing** | e2-medium (2 vCPU, 4 GB RAM) | Standard cost-effective general-purpose compute suitable for containerized Node.js |
| **Containerization** | Docker Engine (Ubuntu 22.04 LTS) | Stateless microservice container deployed via GCE Instance Template startup script |
| **Traffic Management** | Google Cloud Load Balancing | External Application Load Balancer; Layer 7 round-robin routing; HTTP health probes |
| **Horizontal Elasticity** | Compute Engine MIG Autoscaler | Reactive metric rules: Scale-out @ CPU > 70%, Scale-in @ CPU < 30%, 300s cool-down |
| **Synthetic Load Generator** | Grafana k6 (Go-based) | Distributed virtual user (VU) scenarios generating step-function flash-sale traffic curves |
| **Artifact Registry** | GitHub Container Registry (ghcr.io) | CI/CD container image storage with version-pinned automated pulls |

## 7.3 Microservice Endpoints & Stateless Architecture

The MyntraFlash Node.js/Express service exposes six purpose-built endpoints: (1) `GET /` (responsive store interface); (2) `GET /products` (catalog retrieval with simulated database latency); (3) `GET /flash-sale` (time-limited discount items); (4) `GET /api/discount` (simulates computational rule evaluation with configurable matrix calculations); (5) `GET /api/recommend` (simulates feature-lookup and scoring latency); and (6) `GET /health` (synthetic health probe returning HTTP 200 OK for load-balancer vitality checks).

Architectural Statelessness: The application maintains zero local in-memory session state and writes no local disk state. Any VM instance can satisfy any HTTP request interchangeably. This statelessness guarantees that Google Cloud Load Balancing can employ pure round-robin distribution, newly provisioned MIG nodes provide instantaneous traffic relief, and scale-in terminations never disrupt active user sessions.

### 7.4 Proposed Implementation Architecture Diagram



*Figure 2: Architectural design of the proposed MyntraFlash cloud simulation on Google Cloud Platform (GCP).*

# 8. Exercise & Experimental Methodology (Week 3)

## 8.1 Experimental Design Overview

In strict compliance with course requirements, the Week 3 practical exercise executes five controlled experiments measuring response latency (p50, p95, p99), horizontal scaling linearity, autoscale elasticity, fault-recovery resilience, and economic cost:

| | | | |
|---|---|---|---|
| **Exp 1: Baseline** | Single-instance performance limit | 1 GCE VM fixed; 50 sustained VUs; 5 min | p50, p95, p99 latency, req/s, CPU %, Error rate |
| **Exp 2: Scaling** | Horizontal scale comparison (1 vs 2 vs 3 VMs) | Identical 100 VU load; 5 min per run | Latency curves vs instance count; throughput linearity |
| **Exp 3: Elasticity** | Autoscale responsiveness & cooldown | Bounds 1–3 VMs; dynamic 20→500 VU curve | Scale-out lag, scale-in lag, flapping prevention, CPU history |
| **Exp 4: Resilience** | Health probe failover & fault tolerance | 2 VMs; sudden container crash injection | Detection time, connection draining, HTTP error rate |
| **Exp 5: Economics** | Fixed vs elastic cloud billing comparison | 24-hour flash-sale synthetic profile | Total VM-hours, compute cost, disk/LB cost, net ROI |

## 8.2 Flapping Prevention & Cooldown Mechanics

Autoscaling instability ("flapping") occurs when instance additions immediately drop average fleet metrics below scale-in thresholds. Our design enforces stability through a wide hysteresis band (Scale-out at CPU > 70%; Scale-in at CPU < 30%) and a mandatory 300-second (5-minute) cool-down period. With average fleet CPU computed as:

*CPU_fleet = (1 / N) * SUM(CPU_i), adding a second instance to an 80% saturated VM drops fleet average to ~40%, safely residing within the 30%–70% neutral band and preventing premature termination.*

## 8.3 24-Hour Economic Cost Model on GCP: Fixed vs. Elastic Provisioning

Based on verified Google Cloud Platform pricing for the asia-south1 Mumbai region (Pay-As-You-Go, Ubuntu Linux, e2-medium at $0.0335/hr, External Application Load Balancer at $0.025/hr, 30GB Standard Persistent Disk at $0.040/GB/mo), we compare a fixed 3-VM deployment against an elastic 1–3 VM profile over a 24-hour cycle (20 hrs baseline @ 1 VM, 2 hrs ramp @ 2 VMs, 2 hrs peak @ 3 VMs):

| | | | |
|---|---|---|---|
| **Total Compute VM-Hours** | 3 VMs × 24 hrs = 72.0 VM-hours | (20×1) + (2×2) + (2×3) = 30.0 VM-hours | 42.0 VM-hours saved (-58.3%) |
| **GCE Compute Cost (e2-medium)** | 72.0 hrs × $0.0335 = $2.412 | 30.0 hrs × $0.0335 = $1.005 | $1.407 saved on compute |
| **Standard Persistent Disk (30GB)** | 3 disks × 24 hrs = $0.118 | Dynamic disk allocation = $0.049 | $0.069 saved on storage |
| **Cloud Load Balancer Base** | 24 hrs × $0.0250 = $0.600 | 24 hrs × $0.0250 = $0.600 | Fixed baseline infrastructure cost |
| **Data Ingress / Egress (50GB)** | 50 GB × $0.0080 = $0.400 | 50 GB × $0.0080 = $0.400 | Equivalent network payload cost |
| **Cloud Monitoring Telemetry** | $0.000 (Included in GCP tier) | $0.000 (Included in GCP tier) | Zero metric telemetry surcharge |
| **Total 24-Hour Cost** | $3.530 USD (~ ₹295 INR) | $2.054 USD (~ ₹171 INR) | 41.8% Net Overall Cost Reduction |
| **30-Day Extrapolated Cost** | $105.90 USD (~ ₹8,842 INR) | $61.62 USD (~ ₹5,145 INR) | ₹3,697 INR Monthly |

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|---|---|---|---|
| | | | Operational Savings |
| **Student Out-of-Pocket Expense** | ₹0.00 (Covered by $300 credit) | ₹0.00 (Covered by $300 credit) | Zero financial risk for students |

# 9. Team Roles & Project Execution Governance

## 9.1 Team Responsibility Matrix

| | | | | |
|---|---|---|---|---|
| **Myntra Case Study Research & Source Auditing** | Lead | Support | Support | Peer-reviewed citations |
| **Production Architecture Reconstruction** | Lead | — | Support | Mermaid conceptual modeling |
| **MyntraFlash Express Backend API Implementation** | — | Lead | — | GitHub PR code review |
| **Storefront UI Design & Endpoint Integration** | — | Lead | — | Cross-browser testing |
| **Docker Packaging & Multi-Platform ghcr.io Push** | — | Lead | Support | Automated image tagging |
| **GCP Resource Provisioning (VPC, LB, MIG)** | — | — | Lead | gcloud CLI automation |
| **Cloud Monitoring & MIG Autoscaler Tuning** | — | — | Lead | Hysteresis verification |
| **Grafana k6 Load Testing Script Development** | — | — | Lead | Scenario staging & thresholds |
| **Experimental Execution & Telemetry Recording** | Support | Support | Lead | Collaborative lab sessions |
| **Empirical Latency & Cost Data Analysis** | Lead | Support | Support | Statistical aggregation |
| **Week 2 Deliverable Compilation & Formatting** | Lead | Support | Support | Academic report synthesis |
| **Week 4 Consolidated Final Report Preparation** | Lead | Support | Support | Comprehensive submission |
| **Final Project Viva Presentation & Defense** | Shared (33%) | Shared (33%) | Shared (33%) | Joint oral defense |

## 9.2 Week 3 Implementation Timeline & Milestones

| | | | |
|---|---|---|---|
| **Day 1** | App Development & Containerization | Express service code, Dockerfile, GHCR image | Ritul Patel |
| **Day 2** | GCP Infrastructure Provisioning | VPC, Subnet, Firewall, Cloud LB, HTTP Health Check | Divyesh Vagh |
| **Day 3** | MIG Deployment & Autoscale Rules | MIG cluster with startup-script Docker bootstrap | Divyesh Vagh |
| **Day 4** | Experiments 1, 2, and 4 (Static Loads) | Baseline, Horizontal Scaling & Fault-Tolerance JSONs | Divyesh & Utsker |
| **Day 5** | Experiment 3 (Dynamic Flash- | Autoscale CPU time-series & | Divyesh & Ritul |

| | | | |
|---|---|---|---|
| | Sale Autoscale) | instance event logs | |
| **Day 6** | Experiment 5 & Statistical Synthesis | GCP Cloud Billing extraction & latency tables | Utsker Dhameliya |
| **Day 7** | Week 3 Results Documentation | Completed markdown tables and chart artifacts | Entire Team |

## 9.3 Comprehensive Risk Management Matrix

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|---|---|---|---|---|
| **Accidental Cloud Billing Post-Trial** | High | Low | GCP enforces zero auto-charge; set $10 Cloud Billing budget alert; delete MIG post-test | Divyesh |
| **Compute Engine Quota Bottleneck** | Low | Low | Default regional vCPU quota (24+ vCPUs) easily accommodates 3 e2-medium instances (6 vCPUs) | Divyesh |
| **MIG Autoscaler Scale-Out Delay** | Medium | Medium | Set target CPU to 70% with aggressive 150 VU load ramp; pre-cache container images | Divyesh |
| **Client Test Runner Bottleneck (k6 CPU)** | Medium | Low | Execute k6 from dedicated local terminal or adjacent GCE VM; monitor client CPU | Divyesh |
| **Container Image Pull Failures on Boot** | High | Low | Ensure GHCR package visibility is public; embed fallback health script in startup metadata | Ritul |
| **Network Jitter in Latency Measurements** | Low | High | Execute each test run 3 times; report median and 95th percentiles across runs | Utsker |

## 10. Expected Outcomes & Conclusion

### 10.1 Expected Academic & Pedagogical Outcomes

1. Empirical Verification of Horizontal Elasticity: Validating whether doubling GCE instances halves tail latency (p95/p99) under fixed transaction loads, establishing the practical limits of Amdahl's Law in web applications.

2. Practical Autoscale Dynamics: Experiencing first-hand the trade-offs between reactive metric lag, VM provisioning duration, and cooldown stabilization.

3. Autonomous Self-Healing Validation: Demonstrating zero user-perceptible outages during single-instance node crashes via automated load-balancer health check probe deregistration.

4. Quantifiable Cloud Cost Optimization: Providing concrete mathematical and billing evidence that elastic autoscale architectures achieve ~42% operational savings compared to static peak-capacity allocations on Google Cloud Platform.

### 10.2 Simulation Constraints & Technical Limitations

While MyntraFlash effectively models cloud elasticity, several real-world production complexities remain out of scope: (1) Scale: Myntra operates thousands of distributed pods across multi-region clusters, whereas MyntraFlash evaluates 1 to 3 VMs; (2) Data Persistence: MyntraFlash utilizes mock in-memory catalog data, bypassing database replication, distributed locking, and storage I/O bottlenecks; (3) Complex Microservice Topology: Production Myntra coordinates hundreds of interdependent services via Kafka, whereas MyntraFlash encapsulates functions in a single container; and (4) Geographic Latency: Testing from a centralized test runner does not replicate real-world multi-ISP network jitter across pan-Indian mobile networks.

### 10.3 Conclusion

Myntra's engineering evolution demonstrates that scaling for e-commerce flash sales is fundamentally an architectural challenge, not merely a compute-provisioning exercise. Sub-millisecond personalization, 20× flash-sale resilience, and 35% data cost efficiencies were unlocked through the deliberate convergence of cloud-native elasticity, distributed streaming ingestion (Sourcerer & Kafka), near-real-time event computation (Quicksilver & Janus), unified lakehouses (Bifrost & Delta Lake), and ultra-low-latency feature serving (Aerospike).

The MyntraFlash exercise developed in this Week 2 deliverable translates these enterprise principles into an achievable, rigorous, and highly educational laboratory testbed on Google Cloud Platform. By establishing a robust experimental plan and division of team accountability backed by GCP's accessible $300 free trial with zero auto-charge protection, Group 33 is fully prepared to execute the deployment, benchmarking, and cost analysis in Week 3 with zero financial risk.

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